

End-to-End AI Project Report: Titanic Survival Prediction

1. Problem Statement & Objective

The objective of this project is to build a binary classification machine learning model that predicts passenger survival on the Titanic based on features like Age, Gender, Ticket Class, and Fare.

2. Data Preprocessing & Methodology

Data Cleaning: Missing values in 'Age' were imputed using the median, and 'Embarked' using the mode. High extreme outliers in 'Fare' were capped at the 99th percentile.

Feature Engineering: Categorical variables ('Sex', 'Embarked') were transformed using One-Hot Encoding. Non-predictive columns ('Name', 'Ticket', 'Cabin', 'PassengerId') were dropped.

3. Model Performance & Evaluation

Metric	Logistic Regression	Random Forest (Tuned)
Accuracy	81.01%	82.68%
Precision	78.57%	81.16%
Recall	74.32%	75.68%
F1-Score	76.39%	78.32%
ROC-AUC	0.798	0.814

4. Actionable Business Insights

- **Gender Impact:** Female passengers had a significantly higher survival rate (~74%) compared to males (~19%), validating the prioritization of women and children during evacuation.
- **Socio-Economic Class:** Passengers in 1st Class had the highest survival probability, indicating priority access to lifeboats based on socio-economic status.
- **Model Choice:** Hyperparameter-tuned **Random Forest Classifier** outperformed Logistic Regression across all metrics (F1-Score: 0.7832), making it the recommended decision-making model.